

TOPIC LOOKUP — READ FIRST

average power density (S_{av})	POWER R&T (1,1)
angle of reflection (θ_r)	ANGLES (1,2)
critical angle (θ_c)	ANGLES (1,2)
angle of refraction (θ_t , Snell)	ANGLES (1,2)
angle of transmission (θ_t , Snell)	ANGLES (1,2)
tangent of Brewster angle ($\tan \theta_B$)	ANGLES (1,2)
Brewster angle (θ_B)	ANGLES (1,2)
polarization for reflected/transmitted waves (at θ_B)	ANGLES coral note
enclosing a waveguide on both sides w/ length (f_{mnp})	CAVITY RES. (1,3)
large aperture ($d \geq$ several λ)	APERTURE ANT. (2,3)
circular aperture (diam. d)	APERTURE ANT. HPBW tbl. (2,3)
direction of max radiation (θ_{max})	BEAM PARAMS (2,2)

Ê PHASOR TEMPLATES (Air: $\epsilon_r = \mu_r = 1$)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega \sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$$

Direction table $u \in \{x, y, z\}$ = axis perp. to boundary

Inc dir	Inc	Refl	Trans
$+\hat{u}$	$e^{-jk_1 u}$	$e^{+jk_1 u}$	$e^{-jk_2 u}$
$-\hat{u}$	$e^{+jk_1 u}$	$e^{-jk_1 u}$	$e^{+jk_2 u}$

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u} \text{ (V/m)} \quad \text{(incident wave, Med 1)}$$

$$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u} \text{ (V/m)} \quad \text{(reflected wave, Med 1)}$$

$$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u} \text{ (V/m)} \quad \text{(transmitted wave, Med 2)}$$

(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)

Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.

$\hat{e}$ by polarization plug into templates above

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 ($z < 0$) incident side; Med. 2 ($z \geq 0$) transmitted.</i>			
	Normal $\perp$ pol ($\theta_i = 0$)	$\parallel$ pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{\parallel} ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_1}{\eta_2}$	$ \tau_{\perp} ^2 \frac{2\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{\parallel} ^2 \frac{2\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
$R+T$ total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$
Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; nonmag. $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$.			

Intrinsic impedance η_i plug into table above
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$
Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says “ferromagnetic”/iron/ferrite/etc.
Low-loss correction $\eta_c \propto /(\omega \epsilon) \ll 1$
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega \epsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$
For Γ/τ just use $\eta_0/\sqrt{\epsilon_r}$. The j term is for $ \eta_c , \theta_\eta$. See LOSSY MEDIA — 5 CASES.

POWER REFLECTION & TRANSMISSION

Avg power density of plane wave <i>lossless; E_0 peak E-field amp.</i>
$S_{av} = \frac{ E_0 ^2}{2\eta} \text{ (W/m}^2\text{)}$
$\eta = \text{intrinsic impedance of the medium (}\Omega\text{): } \eta = \sqrt{\mu/\epsilon} = \eta_0 \sqrt{\mu_r/\epsilon_r} \xrightarrow{\mu_r=1} \eta_0/\sqrt{\epsilon_r}; \quad \eta_0 = 120\pi \approx 377 \Omega \text{ (free space).}$
% reflected <i>always</i> % $P_r = 100 \Gamma ^2$
% transmitted <i>η-ratio matters!</i> % $P_t = 100 \tau ^2 \eta_1/\eta_2$
Check: % P_r + % $P_t = 100$.

BOUNDARY

η_1, η_2 — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
$\tau = 1 + \Gamma; R = \Gamma ^2$
$k_i = \omega \sqrt{\epsilon_{r,i}}/c$ (rad/m)
α_2 (Np/m), β_2 (rad/m): lossy $z=0$ — boundary plane (m)
$\eta_0 = 120\pi \approx 377 \Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
$n_i = \sqrt{\epsilon_{r,i}}$ — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k +normal
$\perp$: $\mathbf{E} \perp$ this plane
$\parallel$: $\mathbf{E} \parallel$ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m); $a > b$
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
$Z_{TE/TM}$ — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
$\eta = \eta_0/\sqrt{\epsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\epsilon_r}$ — bulk phase vel.
$u_p = u_{p0}/\sqrt{1 - (f_c/f)^2}$
$u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$
$u_p u_g = u_{p0}^2$
$t = L/u_g$ — travel time
$c = 3 \times 10^8$ m/s

POWER & UNITS

% P_r , % P_t — pwr refl./trans. (unitless)
% $P_r = 100 \Gamma ^2$
% $P_t = 100 \tau ^2 \eta_1/\eta_2$
% P_r + % $P_t = 100$
σ (S/m), ϵ (F/m), μ (H/m)
$\epsilon_0 = 8.85 \times 10^{-12}$ F/m
$\mu_0 = 4\pi \times 10^{-7}$ H/m

ANGLES — REFLECTION, TRANSMISSION, CRITICAL, BREWSTER

Reflection angle (Snell, reflection) <i>always</i> $\theta_r = \theta_i$ (from normal)
Transmission/refraction (Snell, refraction) <i>single boundary; angles from normal</i> $\sin \theta_t = \sin \theta_i \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_i \frac{n_1}{n_2}$
Multilayer chain (Snell, chained) <i>stack of slabs 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4</i> $\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical; collapse of Snell chain</i> $\theta_{\text{exit}} = \theta_{\text{incident}}$
Critical angle θ_c <i>TIR when $n_1 > n_2$; no transmitted wave</i> $\sin \theta_c = \frac{n_2}{n_1} = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \Rightarrow \theta_c = \arcsin\left(\frac{n_2}{n_1}\right)$
Brewster angle θ_B ($\parallel$ -pol only) $\Gamma_{\parallel} = 0$; nonmag $\tan \theta_B = \frac{n_2}{n_1} = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}} \Rightarrow \theta_B = \arctan\left(\frac{n_2}{n_1}\right)$
At θ_B : reflected wave is pure $\perp$ -pol ($\parallel$ reflection = 0); transmitted has both. Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients ($\parallel$ vs $\perp$).

LATERAL DISPLACEMENT

Beam offset through N slabs thickness t_i , refracted angle inside slab i Slab-indexed θ_i = angle inside slab i ; t_i = slab thickness (m) $d = \sum_{i=1}^N t_i \tan \theta_i$ The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

EX — HW4 P7 (aperture scaling)

Circular aperture, $\beta = 3^\circ$ at $f = 20$ GHz.
(a) $D = \frac{4\pi}{\beta^2} = \frac{4\pi}{(\pi/60)^2} = \boxed{4584} \rightarrow \mathbf{36.6 \text{ dB}}$
(b) $A_p \times 2$, f same: $D' = 2D = 9168 \rightarrow \boxed{39.6 \text{ dB}}$, $\beta' = \beta/\sqrt{2} = \boxed{2.12^\circ}$
(c) $f \times 2$, A_p same: $D' = 4D = 18336 \rightarrow \boxed{42.6 \text{ dB}}$, $\beta' = \beta/2 = \boxed{1.5^\circ}$

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0; \epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$ $\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$		
Type	$\sigma/\omega \epsilon$	η_c, α, β
Lossless ($\sigma = 0$)	0	$\eta = \eta_0/\sqrt{\epsilon_r}$ exact $\alpha = 0, \beta = \omega \sqrt{\epsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma}{2\omega \epsilon'}\right)$ $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	10^{-2} – 10^2	exact: $(\eta/\sqrt{X}) e^{j\theta_\eta}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f \mu/\sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\epsilon_r} \eta_0$
For Γ/τ : drop j correction; use $\eta_0/\sqrt{\epsilon_r}$. Magnetic: keep μ_r .		
Low-loss quick params $\sigma/(\omega \epsilon) \ll 1, \mu_r = 1$	$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}} \text{ (Np/m)}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}$	
Exact (quasi-cond) $X = \sqrt{1 + (\epsilon''/\epsilon')^2}$	$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1}$ $\theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$	

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$ TM mode $H_z = 0, E_z \neq 0$
Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.
E_x form (TE) <i>read m, n from arguments</i> $E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$
coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.
Identify TE $_{mn}$ vs TM $_{mn}$ <i>cos/sin pattern of E_x is identical for both!</i>
1. Read the problem statement (e.g. “a TE wave” $\rightarrow$ TE).
2. If $m=0$ or $n=0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).
3. Source field $H_z(x, y)$ given $\Rightarrow$ TE. Source field $E_z(x, y)$ given $\Rightarrow$ TM.
TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. So TE $_{10}$ exists but TM $_{10}$ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m , $\tilde{H}$ in A/m , Z in Ω .
TE mode <i>generator: $\tilde{H}_z$</i> $\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$ $\tilde{E}_x = \frac{j\omega \mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$ $\tilde{E}_y = -\frac{j\omega \mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$ $\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \tilde{H}_y = \tilde{E}_x/Z_{TE}$ $Z_{TE} = \eta/\sqrt{1 - (f_c/f)^2}$
TM mode <i>generator: $\tilde{E}_z$</i> $\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$ $\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$ $\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$ $\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \tilde{H}_y = \tilde{E}_x/Z_{TM}$ $Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$
TEM plane wave <i>for reference</i> $\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x/\eta$ $\eta = \sqrt{\mu/\epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A, } k = \omega \sqrt{\mu \epsilon}$
Properties common to TE/TM $f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$ $\beta = k \sqrt{1 - (f_c/f)^2} \text{ (rad/m)}$ $u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c/f)^2}} \text{ (m/s)}$

CUTOFF FREQUENCY

Common cutoffs ($a > b$) f_c formula in Table 8-3	
Mode	f_{mn}
u_{p0}	$c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$)
TE $_{10}$	$f_{10} = u_{p0}/(2a)$ (dominant)
TE $_{01}$	$f_{01} = u_{p0}/(2b)$
TE $_{20}$	$f_{20} = u_{p0}/a = 2f_{10}$
TE $_{11}, \text{TM}_{11}$	$f_{11} = \frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$. In Z_{TE}/Z_{TM} and $u_g, f_c = f_{mn}$ of the excited mode.

CAVITY RESONATOR (guide closed both ends, length d)

Resonant frequencies $m, n, p = \text{mode indices; standing wave in all 3 dims}$
$f_{mnp} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2} \text{ (Hz)}$
TE $_{mnp}$: $p \geq 1$, at least one of $m, n \geq 1$. TM $_{mnp}$: $m, n \geq 1, p \geq 0$.
Dominant TE $_{101}$ when $a \geq d > b$: $f_{101} = \frac{c}{2\sqrt{\epsilon_r}} \sqrt{1/a^2 + 1/d^2}$.

ϵ_r FROM DISPERSION

Given ω, β , mode (m, n) <i>result is unitless</i>
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$ (unitless)
Inputs: ω (rad/s), β (rad/m), a, b (m), $c = 3 \times 10^8$ (m/s), m, n (unitless).

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}, \quad u_p u_g = u_{p0}^2$
$t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower $u_g \Rightarrow$ arrive later.

DIPOLE TYPE BY LENGTH
 l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
l/λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73 \Omega$
= 1	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199 \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$	
Avg power density $l/\lambda < 1/50$; far field $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta$ (W/m ²)	
<i>Equivalent:</i> $S = S_{\text{max}} \sin^2 \theta$ with $S_{\text{max}} = \eta_0 k^2 I_0^2 l^2 / (32\pi^2 R^2)$ at $\theta = \pi/2$.	
l : rod length (m)	I_0 : peak current (A)
R : obs. distance (m)	θ : angle from dipole axis (rad)
$k = 2\pi/\lambda$ (rad/m)	$\eta_0 = 120\pi \approx 377 \Omega$
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)	

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Avg power density any l; far field $S(R, \theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$ (W/m ²)	
<i>At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.</i>	
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$	

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$ $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$						
°	rad	sin θ	cos θ	sin ² θ	cos ² θ	
0	0	0	1	0	1	
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4	
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2	
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4	
90	$\pi/2$	1	0	1	0	
180	π	0	-1	0	1	

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY
 N = # of elements, d = interelement spacing (m)

Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$ $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$	
Normalize $F_a^{\text{norm}} = F_a/N^2$	
HPBW (broadside, uniform) use Nd, not $(N-1)d$ $\beta \approx 0.886 \frac{\lambda}{Nd}$ (rad) = $50.8^\circ \cdot \frac{\lambda}{Nd}$	
Electronic steering (scan to θ_0) replace γ with $\gamma' = kd(\cos \theta - \cos \theta_0)$ $\delta = \frac{2\pi d}{\lambda} \cos \theta_0$ ($\frac{\text{rad}}{\text{elt}}$); $\theta_0 = \frac{\pi}{2}$: broadside, 0: end-fire	
Frequency scan feed-line incr. $\Delta t = n_0 \lambda_0$ $\cos \theta_0 \approx \frac{n_0 \lambda_0}{d} \frac{\Delta f}{f_0}$	
<i>Grating lobes appear if $d > \lambda$. Avoid.</i>	

EX — HW4 P8 (5-elem array, $d=3\lambda/4$)

$N=5$, broadside, equal phase, uniform amp. $kd=2\pi(3/4)=3\pi/2$; $\gamma = \frac{3\pi}{2} \cos \theta$; $F_a^{\text{max}}=25$ at $\theta = \pi/2$	
$F_a^{\text{norm}} = \frac{1}{25} \cdot \frac{\sin^2((15\pi/4) \cos \theta)}{\sin^2((3\pi/4) \cos \theta)}$	
$\beta = 0.886 \frac{\lambda}{Nd} = 0.886 \cdot \frac{4}{15} = \boxed{0.236 \text{ rad} = 13.5^\circ}$	

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c = 3 \times 10^8$ m/s
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377 \Omega$
$S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
a — coef. on θ^2 in Gaussian
λ — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ radiation eff. (unitless); $G = \xi D$
$D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIS / dB

P_t — power transmitted (W)
P_r — power received (W)
G_t — transmit gain (linear)
G_r — receive gain (linear)
$S_r = G_t P_t / (4\pi R^2)$ (W/m ²)
$P_r = G_r S_r (\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB = $\times 2$, 10 dB = $\times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m ²)
$A_e = \lambda^2 D / (4\pi)$ (m ²)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88\lambda/\ell$ (rect/sq.)
$\beta \approx 1.02\lambda/d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd \cos \theta$
$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
$F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.886\lambda/(Nd)$ (broadside)

BEAM PARAMETERS — θ_{max} , β , Ω_p , D

Normalize always start here $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)	
Direction of max radiation θ_{max} angle where F peaks; check 3 ways	
(1) Set $\frac{dF}{d\theta} = 0$; (2) sample F at endpoints + middle; (3)	
Pattern type	θ_{max}
Hertzian, half-wave, dipole, $\sin^k \theta$	$\lambda/4$ $\pi/2$ (broadside)
Linear array (broadside, $\delta = \pi/2$)	0)
Linear array (end-fire, $\delta = 0$ or π)	kd)
Steered array (linear phase $\arccos(\delta\lambda/2\pi d)$)	δ)
Aperture (rect., uniform il-0 boresight lum.)	Arbitrary $F(\theta)$ set $dF/d\theta = 0$ or sample
Beamwidth β at threshold X Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)	
Threshold	$X = 10 \text{ dB}/10$ θ_X
HPBW (−3 dB)	0.5 $\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25 $\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1 $\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X $F(\theta_X) = X$

Pattern solid angle Ω_p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$ (sr)	
<i>No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$. θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.</i>	

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F=1$)	$4\pi \approx 12.57$
Hertzian $\sin^2 \theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_1^2/2 / \ln 2$
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$
Other (use integral)	numerical

TI-36X: [2nd] [d/dx] for $\int_0^\pi$, RAD mode, multiply by 2π .

Directivity D always $D = 4\pi/\Omega_p$; common shortcuts below $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$	
WORD TRAP: “direction” = θ_{max} (rad); “directivity” = D (unitless).	
Gain (if asked) ξ = radiation efficiency (unitless), $0 < \xi \leq 1$ $G = \xi D$ (unitless) $G_{\text{dB}} = 10 \log_{10} G$ (dB)	
Lossless / ideal antenna $\Rightarrow G = D$.	
$\xi = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}}$; $R_{\text{rad}} = 2P_{\text{rad}}/I_0^2$ (Ω); $R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}$.	

COMMON DIPOLE PARAMETERS

Lossless ($\xi=1$): $G=D$.				
Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199
Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.				

ANTENNA AREAS (A_e , A_p)

Effective area antenna's RX cross section; D from BEAM PARAMS or table above $A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)	
Physical cross section wire dipole, diameter d; l see table above $A_p = l d$ (m ² , any dipole)	
Inverse: D from A_e $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)	
<i>Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).</i>	

FRIS TRANSMISSION FORMULA

For each antenna's G : if given by NAME (“half-wave dipole” etc.) $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\rightarrow$ use as given (convert dB $\rightarrow$ linear).
Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$
 $S_r = \frac{G_t P_t}{4\pi R^2}$ (W/m²); $P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$ (W)

Equivalent: $P_r = S_r A_e$, r.
Solve for R range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$$

Equivalent path using A_e recv. area form
 $P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$, $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$
Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned
 $P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.

3. Get G_t, G_r by antenna type:
- named dipole $\rightarrow$ COMMON DIPOLE PARAMS ($G = D$ lossless)
 - aperture / dish $\rightarrow$ APERTURE ANTENNAS ($G = D \approx 4\pi A_p/\lambda^2$)
 - arbitrary pattern $F(\theta) \rightarrow$ BEAM PARAMS ($G = \xi \cdot 4\pi/\Omega_p$)
 - given in dB $\rightarrow$ dB $\leftrightarrow$ LINEAR table
4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity ($G, D, P_r/P_t, \text{SNR}, \dots$):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver $P_N = k_B T_{\text{sys}} B$ (W)	
$k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).	
Signal-to-noise ratio $\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$	

APERTURE ANTENNAS (horn, dish; large aperture, $d \geq$ several λ)

A_p physical aperture area (m²); A_e effective area (m², $= \lambda^2 D / 4\pi$); β_{xz}, β_{yz} HPBW's in two planes (rad); ℓ, d aperture side / diameter (m).

Far-field req.: $R \geq 2d^2/\lambda$ (d = largest aperture dim.).

Pattern (rect., uniform illum.) peak at $\theta = 0$ (broadside)

$$F(\theta) = \text{sinc}^2 \left(\frac{\pi \ell_x \sin \theta}{\lambda} \right); \quad S_0 = \frac{E_0^2 A_p}{2\eta_0 \lambda^2 R^2}$$

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths any aperture

$$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and D by aperture shape uniform illum.; β in rad

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= \frac{4\pi \ell_x \ell_y}{\lambda^2}$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx \frac{4\pi}{\beta^2} = \frac{\pi^2 d^2}{\lambda^2}$

Scaling (any ratio) $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$
 $D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$
 $\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$
If a parameter is unchanged, its ratio = 1. Examples: double area $\rightarrow D \times 2$, $\beta \times 1/\sqrt{2}$; double freq $\rightarrow D \times 4$, $\beta \times 1/2$.